Generator Identifiability, Operational Observables, and Physical-Selection Limits

A Technical Supplement to Hierarchical Shell Manifold Theory

Vince Garrett
Hierarchical Shell Manifold Theory Research Program

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Technical research manuscript—not a derivation of a fundamental physical law

Abstract

Hierarchical Shell Manifold Theory (HSMT) uses a graded Hilbert-space architecture and a reference operator to organize candidate effective dynamics, but its frozen mathematical core does not select a physical evolution generator or an operational measurement structure. We consolidate four finite-dimensional generator-identifiability investigations (PS-1A–PS-1D) and an ancillary multi-flow nonselection result (MT2). A two-grade transition model exhibits exact sign ambiguities under population-only observations, while coherent observables recover signed parameters relative to calibrated conventions. An interacting grade–internal model contains a genuine internal frequency that is invisible not only to single-time grade observables, but to arbitrary multi-time protocols confined to grade-only operations, with the initial preparation held fixed. Access to internal coherence removes that ambiguity. We distinguish observational equivalence, representation equivalence, and dynamical selection; characterize nested mathematical observable algebras; and construct a symmetry obstruction to choosing a preferred internal generator from the frozen core. These are conditional mathematical results for declared laboratories, not evidence that the universe realizes those Hamiltonians. Neither a canonical physical observable algebra nor the HSMT physical-selection principle is derived.

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1 Introduction and scope

The HSMT program separates its foundational mathematical organization from an as-yet-underived physical generator. In schematic form its fixed data are

$$\mathcal{C} = (\mathcal{H}, \{P_N\}, D), \qquad \mathcal{H} = \bigoplus_N \mathcal{H}_N, \tag{1.1}$$

where the P_N are mutually orthogonal grade projectors and D is a self-adjoint reference operator. Neither the label N nor the grade decomposition is, by itself, a claim about physical spacetime dimension. Self-adjointness of D supplies a mathematical one-parameter unitary group but does not establish D as the physical Hamiltonian; compare Stone's theorem [1].

An observationally identified generator and a fundamentally *selected* generator answer different questions. In the first, an allowed set of preparations, controls, observations, and recorded time-dependent statistics is given. In the second, one seeks independently justified reasons why a particular physical law should be realized. The latter is not obtained merely by estimating a Hamiltonian successfully. This supplement makes that distinction formal using fully specified finite-dimensional examples, alongside standard background on quantum information, system identification and reduced dynamics [2, 4, 3].

Claims and limits. The results below establish exact algebraic statements within the declared models. They do not derive quantum dynamics from the HSMT core, a spacetime metric, a canonical experimental apparatus, a mathematical condensate, or a selection law. Previously reported numerical spot checks are supplementary to the proofs and are not treated as independent experimental evidence.

2 Definitions: observational and representation equivalence

Fix a finite-dimensional Hilbert space $\mathcal{H}$, a family of admissible self-adjoint Hamiltonians $\mathfrak{G}$, and a collection $\mathfrak{E}$ of allowed experimental protocols. Each protocol specifies its preparation, controls

and interventions, measurements, their order and times, and whether external phase references are calibrated. For $H \in \mathfrak{G}$, let $p_H(r \mid e)$ denote its probability for recorded outcome string r under protocol e.

Definition 2.1 (Operational equivalence)**.** *Two Hamiltonians are equivalent for an experimental class $\mathfrak{E}$ if*

$$H \sim_{\mathfrak{E}} H' \quad \Longleftrightarrow \quad p_H(r \mid e) = p_{H'}(r \mid e) \text{ for all } e \in \mathfrak{E} \text{ and } r. \tag{2.1}$$

Definition 2.2 (Representation equivalence)**.** *An equivalence of representations is a simultaneous change $H \mapsto VHV^\dagger$, $\rho \mapsto V\rho V^\dagger$, and $M \mapsto VMV^\dagger$ (and likewise for controls), for a fixed unitary V. It leaves all Born-rule statistics unchanged. Such a transformation must not be confused with changing H while holding the experimentally calibrated apparatus fixed.*

An additive scalar $c\mathbb{I}$ is invisible to ordinary closed-system channel tomography because it only contributes a global phase. Accordingly, identification of a generator generally means identification *modulo* specified observational and representation equivalences. The size of an equivalence class depends on the experimental permissions; restricting observables is not the same as imposing a physical superselection rule.

3 PS-1A: an exactly solvable two-grade model

3.1 Transition probabilities and sign ambiguities

Consider $\mathcal{H} = \mathbb{C}^2_{\text{grade}} \otimes \mathbb{C}^2_{\text{int}}$ and

$$H_A(g, \Delta) = \begin{pmatrix} 0 & g \\ g & \Delta \end{pmatrix} \otimes \mathbb{I}_{\text{int}}, \qquad g, \Delta \in \mathbb{R}. \tag{3.1}$$

The internal factor is inert in this benchmark. Let $|0\rangle$ and $|1\rangle$ label the two grades, and set $\Omega = \sqrt{\Delta^2 + 4g^2}$.

Proposition 3.1 (Grade-transfer law)**.** *Starting in $|0\rangle$, the probability of grade-1 occupancy is*

$$p_{0\to1}(t) = \frac{4g^2}{\Omega^2} \sin^2\!\left(\frac{\Omega t}{2}\right). \tag{3.2}$$

For $g \neq 0$, the full nonconstant time trace determines $|g|$ and $|\Delta|$, but neither sign. For $g = 0$ the transfer trace vanishes identically and does not determine Δ.

Proof. Subtract $\Delta\mathbb{I}/2$ from the two-level Hamiltonian. In the convention $Z|0\rangle = |0\rangle$, the traceless part equals $gX - \Delta Z/2$. Its square is $\Omega^2\mathbb{I}/4$, so the off-diagonal unitary amplitude is $-2ig\sin(\Omega t/2)/\Omega$. Taking the modulus squared gives (3.2). Both amplitude and frequency determine g^2 and Δ^2 for $g \neq 0$. The exceptional case follows directly. □

The sign of g is changed by $S = Z_{\text{grade}} \otimes \mathbb{I}_{\text{int}}$: $H_A(-g, \Delta) = SH_A(g, \Delta)S^\dagger$. This is a representation equivalence when preparations and measurements transform accordingly. By contrast, the inability to determine the sign of Δ from *population measurements alone* is an insufficiency of the particular experiment; coherent phase-sensitive measurements can resolve it in a fixed reference convention.

3.2 Coherent reconstruction and corrected sign convention

For the analytically convenient traceless representative

$$\widehat{H}_A = gX + \frac{\delta}{2}Z, \tag{3.3}$$

where δ is the coefficient-defined signed detuning, prepare $|0\rangle$ and measure X and Y in the calibrated Pauli convention $XY = \mathrm{i}Z$. The Heisenberg equations give

$$\left.\frac{d\langle Y\rangle}{dt}\right|_0 = -2g, \qquad \left.\frac{d^2\langle X\rangle}{dt^2}\right|_0 = 2g\delta. \tag{3.4}$$

Thus, for $g \neq 0$,

$$g = -\frac{1}{2}\langle \dot{Y}\rangle_0, \qquad \delta = \frac{\langle \ddot{X}\rangle_0}{2g}. \tag{3.5}$$

Convention note. The block Hamiltonian (3.1), after subtraction of $\Delta\mathbb{I}/2$, has $\delta = -\Delta$. Hence the sign of the $\langle X\rangle$ second derivative changes if the symbol Δ is retained for the block-diagonal offset rather than the Pauli-Z coefficient. This note corrects a sign-convention ambiguity in earlier research relays. The physical identification statements are unaffected once one convention is fixed.

Remark 3.2. *Continuous-time, informationally complete closed-system channel tomography determines the commutator superoperator $-\mathrm{i}[H,\cdot]$, and hence a finite-dimensional H up to an additive scalar. This claim assumes known time calibration, sufficient state preparation and measurement access, and the declared unitary model. It is not a claim about a single isolated snapshot or an infinite-dimensional inverse problem.*

4 PS-1B: hidden internal dynamics

4.1 An interacting four-parameter generator

Now use the genuinely interacting family

$$H(g,\Delta,\lambda,\omega) = gX_g \otimes \mathbb{I}_i + \frac{\Delta}{2}Z_g \otimes \mathbb{I}_i + \lambda Z_g \otimes Z_i + \omega \mathbb{I}_g \otimes Z_i. \tag{4.1}$$

All parameters are real. Write

$$H = H_0 + \omega Q, \qquad Q = \mathbb{I}_g \otimes Z_i, \qquad [H_0, Q] = 0. \tag{4.2}$$

The interaction parameter λ changes grade evolution conditionally on the hidden internal state, whereas ω induces relative internal phases.

Theorem 4.1 (Exact reduced-grade nonidentifiability)**.** *For every fixed initial joint density matrix ρ independent of ω, every time t, and any $\omega_1, \omega_2 \in \mathbb{R}$,*

$$\mathrm{Tr}_i\big(U_{\omega_1}(t)\rho U_{\omega_1}(t)^\dagger\big) = \mathrm{Tr}_i\big(U_{\omega_2}(t)\rho U_{\omega_2}(t)^\dagger\big). \tag{4.3}$$

Consequently all single-time grade-only POVM statistics coincide for the two generators.

Proof. Commutation gives $U_\omega(t) = V_\omega(t)U_0(t)$, where $V_\omega(t) = \mathbb{I}_g \otimes \exp(-\mathrm{i}\omega t Z_i)$. The partial trace is invariant under conjugation by a unitary acting only on the traced-out factor. This proves (4.3) without requiring a factorized initial state. □

Theorem 4.2 (Multi-time grade-only no-go)**.** *Assume the initial joint state is held fixed independently of ω, and that all experimental operations between evolution intervals have Kraus operators $M_{r\alpha} \otimes \mathbb{I}_i$, including conditioned grade-only controls and readouts. Under these restrictions, the probability of every finite grade-only outcome record is independent of ω.*

Proof. Each $M_{r\alpha} \otimes \mathbb{I}_i$ commutes with every $V_\omega(t)$. Likewise $U_0(t)$ commutes with $V_\omega(t)$ because $[H_0, Q] = 0$. In the unnormalized conditioned state for any outcome record, the internal rotations may therefore be commuted to the outside of the ordered product of evolution and Kraus operators. For each Kraus branch they combine to a terminal internal unitary $V_\omega(T)$, where T is the total elapsed evolution time. The trace of each conditioned branch is unchanged by that unitary, and summation over Kraus branches completes the proof. □

Remark 4.3 (Scope of the theorem)**.** *The theorem does not cover intervention on the internal subsystem, an ω-dependent preparation, unmodeled coupling to an external internal-phase reference, or replacement of the Hamiltonian by a noncommuting model. It also does not establish the existence of a fundamental superselection rule against such interventions.*

4.2 How internal coherence exposes hidden parameters

Prepare an internal eigenstate of Z_i with eigenvalue $z = \pm 1$. Up to an internal scalar, the grade Hamiltonian becomes

$$H_z = gX_g + \left(\frac{\Delta}{2} + \lambda z\right) Z_g, \qquad \Delta_z = \Delta + 2\lambda z. \tag{4.4}$$

Thus grade populations distinguish conditional *magnitudes* of detunings, and coherent grade tomography can distinguish signed Δ_z given a phase reference. For two calibrated conditional detunings,

$$\Delta = \frac{\Delta_+ + \Delta_-}{2}, \qquad \lambda = \frac{\Delta_+ - \Delta_-}{4}. \tag{4.5}$$

For a more direct measurement, prepare the internal state $|+X_i\rangle$ and the grade in a Z_g eigenstate $z_g = \pm 1$. Using the Pauli convention $\dot{O} = \mathrm{i}[H, O]$,

$$\frac{d}{dt}\langle \mathbb{I}_g \otimes Y_i \rangle \Big|_{t=0} = 2(\omega + \lambda z_g). \tag{4.6}$$

If s_+ and s_- are the slopes under the two grade preparations, then

$$\omega = \frac{s_+ + s_-}{4}, \qquad \lambda = \frac{s_+ - s_-}{4}. \tag{4.7}$$

The result is exact for the chosen finite-dimensional Hamiltonian and calibrated readout; in particular, the initial internal readout changes the operational experiment class.

5 PS-1C: mathematical observable algebras versus physical access

For the frozen core (1.1), three natural von Neumann algebras are

$$\mathcal{A}_{\text{grade}} = W^*(\{P_N\}), \qquad \mathcal{A}_{\text{struct}} = W^*(\{P_N\}, D), \qquad \mathcal{A}_{\text{full}} = \mathcal{B}(\mathcal{H}). \tag{5.1}$$

They satisfy $\mathcal{A}_{\text{grade}} \subseteq \mathcal{A}_{\text{struct}} \subseteq \mathcal{A}_{\text{full}}$. For unbounded D, $W^*(\{P_N\}, D)$ means the von Neumann algebra generated by the projectors and spectral projections of D; it does not mean adjoining an unbounded operator as a bounded generator. None of these mathematical definitions alone asserts an implementable laboratory.

Proposition 5.1 (Nonuniqueness of an invariant measurement algebra)**.** *For $\mathcal{H} = \mathbb{C}^2_g \otimes \mathbb{C}^2_i$, let $P_\pm = (\mathbb{I}_g \pm Z_g) \otimes \mathbb{I}_i/2$ and $D = Z_g \otimes \mathbb{I}_i$. The fixed data are invariant under each $V = \mathbb{I}_g \otimes U_i$ with $U_i \in U(2)$. Both*

$$\mathcal{A}_1 = \mathcal{B}(\mathbb{C}^2_g) \otimes \mathbb{I}_i, \qquad \mathcal{A}_2 = \mathcal{B}(\mathbb{C}^2_g \otimes \mathbb{C}^2_i) \tag{5.2}$$

are invariant as algebras under this automorphism group, yet $\mathcal{A}_1 \neq \mathcal{A}_2$. Thus automorphism invariance of a candidate observable algebra does not select a unique physically accessible algebra.

Proof. Conjugation by $\mathbb{I}_g \otimes U_i$ fixes every element of $\mathcal{A}_1$ and maps $\mathcal{A}_2$ onto itself. The latter contains internal non-scalar observables absent from $\mathcal{A}_1$. Both therefore pass the stated covariance test, which cannot select between them. □

For this example $\mathcal{A}_{\text{struct}} = \mathcal{A}_{\text{grade}} = \text{span}\{\mathbb{I}_g, Z_g\} \otimes \mathbb{I}_i$. The intermediate algebra $\mathcal{A}_1$ includes grade coherence, so it is strictly larger than $\mathcal{A}_{\text{grade}}$. The fact that $\mathcal{A}_1$ and $\mathcal{A}_2$ are mathematically available does not settle which POVMs, preparations or control operations nature or a given apparatus permits. Physical accessibility is an additional operational assertion.

6 PS-1D: capability-dependent identification

We compare three nested types of experimental capability. The nesting concerns *access*, not necessarily one specific set of initial states.

6.1 Class A: grade-only operations

Take the internal reference preparation $\rho_i = \mathbb{I}_i/2$, with no internal preparations, measurements or controls. The reduced grade dynamics is the equal mixture of the two conditional detuning sectors $z = \pm 1$. The Hamiltonians with λ and $-\lambda$ exchange these sectors, hence produce the same grade-only statistics. Theorem 4.2 also makes ω invisible to every permitted multi-time grade-only protocol. In this declared class, the sign of λ and the value of ω are not identifiable. Other parameters may be recovered in generic nondegenerate cases, but no global injectivity theorem is claimed.

6.2 Class B: controlled internal Z preparations

Now allow separate preparation of the internal sectors $z = \pm 1$, with coherent grade preparations and tomography, but without coherent internal readout. The conditional detunings $\Delta_\pm = \Delta \pm 2\lambda$ can be reconstructed relative to a calibrated grade reference, giving (4.5). The internal frequency

Table 1: Operational identifiability in the declared four-parameter family. Statements concerning signed detunings presume phase calibration and generic parameter values.

Class	Allowed access	Exact boundaries / available recovery
A	Grade-only operations; internal initial state $\mathbb{I}/2$	ω invisible; $\lambda \mapsto -\lambda$ invisible
B	Internal Z_i preparation; coherent grade readout	Signed $\Delta_\pm$ recoverable with calibration; ω invisible
C	Coherent internal preparation and readout	(ω, λ) from two initial slopes

ω remains invisible: in each selected internal eigensector it contributes only a scalar phase, and grade-only protocols cannot compare such phases. Population-only measurements leave further sign ambiguities because they see $\Delta_\pm^2$.

6.3 Class C: internal coherent access

Permit $|+X_i\rangle$ preparation and Y_i readout. The two slopes (4.6) reconstruct (ω, λ) as in (4.7). Full process access can in principle identify the total finite-dimensional Hamiltonian modulo a scalar and agreed representation conventions. This is not a demonstration that full process access follows from HSMT or that the parameters are fundamental.

Remark 6.1 (Gauge, limited access, superselection)**.** *One must distinguish (i) a simultaneous change of mathematical representation, (ii) two fixed-apparatus Hamiltonians that an access-limited observer cannot distinguish, and (iii) a fundamental physical prohibition on an operation. Our finite model demonstrates (i) and (ii), but does not derive (iii).*

7 MT2: why a second mathematical flow is not a selector

The reference self-adjoint operator D generates $U_D(s) = \exp(-\mathrm{i}sD)$ and the functional calculus permits $\exp[-\mathrm{i}sf(D)]$ for suitable real f. The grade projectors additionally permit diagonal grade-phase generators $K = \sum_N k_N P_N$ under appropriate boundedness or domain assumptions. These are mathematical constructions; none is automatically a second physical time.

In the finite example $\mathcal{H} = \mathbb{C}_g^2 \otimes \mathbb{C}_i^2$, take $D = Z_g \otimes \mathbb{I}_i$ and the same grade projectors as above. Each

$$K_X = \mathbb{I}_g \otimes X_i, \qquad K_Y = \mathbb{I}_g \otimes Y_i, \qquad K_Z = \mathbb{I}_g \otimes Z_i \tag{7.1}$$

commutes with D and preserves the grades. Nevertheless, the K's need not commute with one another. Inner automorphisms $\mathbb{I}_g \otimes U_i$ preserve the frozen data and rotate these internal directions. An additional generator invariant under all such automorphisms must have the form

$$K = a\mathbb{I} + bD, \qquad a, b \in \mathbb{R}, \tag{7.2}$$

where the restriction uses the two-grade decomposition and Schur's lemma. Such a flow adds no independently preferred internal direction. More generally, $[H_t, H_s] = 0$ can restrict one generator *once the other is independently given*, but it does not choose a distinguished pair. For time-dependent multi-parameter evolution, consistency requires the corresponding flatness equation with the correctly specified evolution conventions; it is not evidence of extra physical time dimensions.

8 Selection versus identification: the remaining boundary

Let $\mathfrak{G}(\mathcal{C})$ be an admissible family of candidate generators compatible with an HSMT realization and let $\mathfrak{E}$ be the actually justified operational class. The physically meaningful object accessible from predictions is the quotient

$$\mathfrak{G}(\mathcal{C})/\sim_{\mathfrak{E}}, \tag{8.1}$$

subject also to legitimate representation equivalences. A set of independently obtained data D can further restrict compatible classes. Even if only one class remains, it does not follow that the framework has *derived why* that class is fundamental. The distinction can be recorded as follows:

$$\boxed{\begin{array}{c}\text{frozen core} \longrightarrow \text{admissible generators} \\ \longrightarrow \text{operational equivalence classes} \\ \longrightarrow \text{independent physical evidence} \\ \longrightarrow \text{empirical discrimination}\end{array}} \tag{8.2}$$

The box deliberately does not end in a logically entailed fundamental selector. A future HSMT selection principle would need independent mathematical or physical justification, exclude non-equivalent predictions without tuned post hoc assumptions, and avoid assuming spacetime or causal structures it is intended to derive.

9 Limitations and scientific status

1. **Finite laboratories only.** The explicit models have dimensions four and do not establish the corresponding claims for unrestricted infinite-dimensional HSMT dynamics.
2. **No physical generator selected.** Self-adjointness and norm preservation characterize admissible evolution, not its physical origin.
3. **No canonical operational algebra.** Grade projectors, D, and mathematical operator algebras do not identify physically implementable preparations and controls.
4. **No new physical time or branching postulate.** The MT2 discussion is an obstruction to inference from mathematically available commuting flows, not an endorsement of a second physical time.
5. **No experimental validation.** Matrix-exponential checks support internal consistency only. They are not observations of an HSMT-specific phenomenon in nature.
6. **No fundamental condensate derived.** The spatiotemporally ubiquitous mathematical-condensate and branching hypotheses are outside the claims established by these identifiability proofs.

10 Conclusion

Within explicit finite-dimensional graded quantum laboratories, the physical-generator identification problem has exact operational obstructions. Population traces hide parameter signs; restricted grade experiments can conceal a true internal frequency even over arbitrarily many measurement

times; and additional coherent operations can reveal otherwise inaccessible dynamics. A graded Hilbert space and self-adjoint reference operator provide natural mathematical structures but do not uniquely specify an experimentally accessible measurement algebra or an independent second flow. The resulting substantive distinction is between *reconstructing a Hamiltonian conditional on a model and apparatus* and *deriving a fundamental physical-selection principle*. The former is tractable in the laboratories studied here. The latter remains open for HSMT.

Research provenance and disclosure

This manuscript consolidates internally developed research benchmarks denoted PS-1A, PS-1B, PS-1C, PS-1D, and MT2. Their role is theorem-based finite-model analysis, not external experimental confirmation. Historical conversational agreement between research assistants is not treated as independent peer review. The signed-detuning convention has been normalized explicitly in Eqs. (3.3)–(3.5); any future manuscript comparison should preserve this correction.

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